

Sheetal Dental Clinic Booking and Management Platform

Academic Project Report (Chapters 1, 2 & 3)

CHAPTER 1: INTRODUCTION

1.1 Background

In the contemporary healthcare landscape, digital systems have shifted from being optional amenities to essential operational infrastructure. Sheetal Dental Clinic, a provider of comprehensive cosmetic, restorative, and general dentistry, relies on creating a seamless and gentle experience for patients from their first point of contact. Historically, patient scheduling and consultation preparation were conducted via manual telephone interactions or paper logs, introducing operational friction and administrative overhead. By deploying an advanced, web-based appointment booking and patient assessment portal, the clinic can digitize and optimize its scheduling workflow. Utilizing modern frameworks like Next.js, Clerk authentication, and MongoDB, this platform automates the intake pipeline, provides interactive self-assessment tools, and dispatches automated SMS confirmation alerts to improve patient throughput and satisfaction.

1.2 Problem Statement

Sheetal Dental Clinic's traditional appointment booking process is highly reliant on manual scheduling, leading to patient phone wait times, front-desk scheduling conflicts, and human recording errors. Furthermore, patients lack a transparent mechanism to estimate treatment costs beforehand or visualize treatment outcomes (such as orthodontic before-and-after results), which contributes to hesitation in booking procedures. The absence of a real-time patient notification loop and an easily accessible patient history portal increases appointment no-shows and strains administrative resources during peak clinic hours.

1.3 Purpose of Project

Issues with the Existing System

- **Inconvenient Booking Channels:** Patients must call during clinic hours, leading to missed booking opportunities during weekends or off-hours.
- **Lack of Cost Transparency:** Administrative staff spend significant time explaining pricing variations for multiple dental treatments over the phone.
- **Manual Record Management:** Paper-based or local logs make it difficult for doctors to retrieve patient history, schedule dates, or add treatment notes instantly.
- **No Automated Reminders:** The manual dispatch of SMS/calls for appointment confirmation is tedious and prone to human omission, resulting in higher cancellation and no-show rates.

How the Proposed System Solves These Problems

- **24/7 Digital Intake Portal:** Patients can request appointments securely at any time using their mobile device or desktop.

- **Interactive Diagnostic Tools:** The integration of a Smile Quiz, Pricing Calculator, and before-and-after Smile Slider lets users self-assess their dental needs and estimate costs prior to checking in.
- **Centralized Admin Dashboard:** The clinic staff can filter, search, reschedule, and delete appointments digitally, while appending persistent doctor notes directly to patient records in a MongoDB collection.
- **Automated SMS Dispatch System:** The system integrates Twilio API triggers that automatically send real-time confirmation texts, reschedule alerts, or cancellation warnings directly to the patient's phone.

1.4 Objectives of the Project

- **To design and implement** a responsive, user-friendly frontend interface that simplifies dental appointment scheduling for patients.
- **To integrate secure, role-based authentication** using Clerk API to separate patient and administrative view rights.
- **To build a dynamic clinic database** with MongoDB to store, retrieve, update, and delete patient appointment metadata securely.
- **To deploy interactive patient tools**, specifically a dental Treatment Cost Calculator, an diagnostic Smile Quiz, and an orthodontic Smile Slider to increase online booking conversions.
- **To establish an automated notification gateway** using Twilio SMS to verify bookings, announce scheduling times, and communicate doctor instructions.

1.5 Scope of the Project

1.5.1 Features (Currently Delivering)

- **Clerk Authentication Engine:** Restricts dashboard views based on custom user roles (e.g., patient vs. admin).
- **Interactive Patient Landing Page:** Built with custom Tailwind CSS v4 styling, fluid scroll effects, and dark/light theme options.
- **Three Patient Utility Tools:**
 1. **Smile Slider:** A dual-image drag-slider illustrating before-and-after treatment outcomes.
 2. **Smile Quiz:** A multi-step React form analyzing dental complaints and recommending treatments.
 3. **Pricing Calculator:** React-driven estimator listing cost projections for selected dental codes.
- **MongoDB Database Backend:** Handles CRUD operations via REST API endpoints for booking management.
- **SMS Gateway:** Automated Twilio SMS generation triggered when appointments are created, confirmed, rescheduled, or cancelled.
- **Admin Control Center:** Allows the doctor to confirm bookings, designate dates/times, and write treatment directions.

1.5.2 Exclusions (Future Scope)

- **Payment Gateway Integration:** The current application calculates pricing but does not process credit cards or UPI payments online; payments are conducted physically at the clinic.
- **Automated Doctor Calendar Sync:** The admin manually sets the date and time of appointments on confirmation rather than checking an integrated, real-time Google Calendar/Outlook sync of the doctors.
- **Teledentistry Video Consultations:** Direct video calls between the dentist and patients are not supported in this version.
- **Electronic Health Records (EHR) Storage:** Uploading and parsing raw dental X-rays or extensive historical medical records is currently excluded.

CHAPTER 2: LITERATURE SURVEY

2.1 System Name 1: Tend Dental

Website: <https://www.hellotend.com>

Tend is a modern, technology-focused dental care provider based in the United States. It offers an online appointment scheduling platform that enables users to book slots at physical clinics, fill out medical intake forms, select specific dental practitioners, and check insurance co-pays in advance. The platform emphasizes transparency and premium customer experience.

Feature List

- Real-time calendar availability selection.
- Integration with dental insurance systems for real-time co-pay estimation.
- Direct selection of individual dentists with bios.
- Secure patient login and digital intake forms.
- SMS and Email appointment reminders.

Technology Overview

- **Frontend:** React, Next.js, styled-components.
- **Backend:** Node.js API services hosted on AWS Cloud.
- **Database:** Relational PostgreSQL database for patient and clinical logs.
- **Third-Party Integrations:** Dental practice management software (PMS) APIs, Twilio SMS.

2.2 System Name 2: Clove Dental

Website: <https://clovedental.in>

Clove Dental is India's largest network of dental clinics, operating hundreds of clinics across multiple states. Their website provides clinic finders, service descriptions, pricing overviews, and appointment inquiry forms. It focuses heavily on geographical distribution and marketing individual treatments.

Feature List

- Clinic locator with Google Maps integration.
- Appointment request form (lead generation style).
- Comprehensive service categories with text explanations.

- Patient reviews and testimonial highlights.
- Integration with backend CRM to routing leads to local clinics.

Technology Overview

- **Frontend:** HTML5, CSS3, JavaScript, jQuery.
- **Backend:** PHP / WordPress CMS.
- **Database:** MySQL database hosting content and form submissions.
- **Third-Party Integrations:** Localized SMS gateways, CRM lead management integration.

2.3 Bibliography

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2. Jon Duckett, *HTML and CSS: Design and Build Websites*, John Wiley & Sons, 2011.
3. Jon Duckett, *JavaScript and jQuery: Interactive Front-End Web Development*, John Wiley & Sons, 2014.
4. Ian Sommerville, *Software Engineering*, 10th Edition, Pearson Education, 2015.
5. Next.js Documentation, *Next.js App Router Paradigm*, Vercel, 2025.
6. MongoDB Manual, *MongoDB Driver CRUD Operations*, MongoDB Inc., 2026.

CHAPTER 3: SYSTEM REQUIREMENTS & DESIGN

3.1 System Modules

- 1. Authentication & Role Switcher Module:** Utilizing Clerk API, this module logs in users, handles signup flows, and extracts metadata roles. For development testing, a role switcher endpoint `/api/user/toggle-role` allows switching metadata between "patient" and "admin" roles.
- 2. Landing Page & Interactive Tools Module:** Houses the patient intake tools:
 - **Smile Quiz:** Multi-step React form analyzing dental complaints and recommending treatments.
 - **Smile Slider:** HTML/CSS slider visualizing orthodontic progress.
 - **Pricing Calculator:** React-driven estimator listing cost projections for selected dental codes.
- 3. Patient Appointment Management Module:** Enables authenticated patients to fill the appointment request form. It displays their bookings dynamically, letting them submit cancellation requests (`status: cancel_requested`) or choose new rescheduling options (`status: reschedule_requested`).
- 4. Admin Booking Controller Module:** A secure panel at `/admin` displaying all clinic appointments. Administrators can search bookings, delete outdated entries, input appointment dates and times, write persistent doctor instructions, and confirm requests (`status: confirmed`).
- 5. SMS Notification Engine:** A server-side utility connected to the Twilio REST API. It is triggered by API actions to format and transmit customized texts when appointments are requested, updated, scheduled, or cancelled.

3.2 Hardware and Software Requirements

3.2.1 Hardware Requirements

Component	Specification
Client / Developer Workstation	Intel Core i3 / Ryzen 3 or higher, 8 GB RAM, 120 GB SSD free space
Server (Cloud Hosting)	Vercel Serverless Environment (Node.js runtime)
Database Server	MongoDB Atlas Cloud instance

3.2.2 Software Requirements

Category	Specification
Frontend	Next.js 16 (React 19 framework), Tailwind CSS v4, Lucide React, Framer Motion

Category	Specification
Backend	Next.js API routes (Node.js runtime)
Database	MongoDB (NoSQL Document Store)
APIs / Integrations	Clerk Auth SDK (Authentication), Twilio SDK (SMS)
IDE & Utilities	VS Code, Git, Chrome Developer Tools

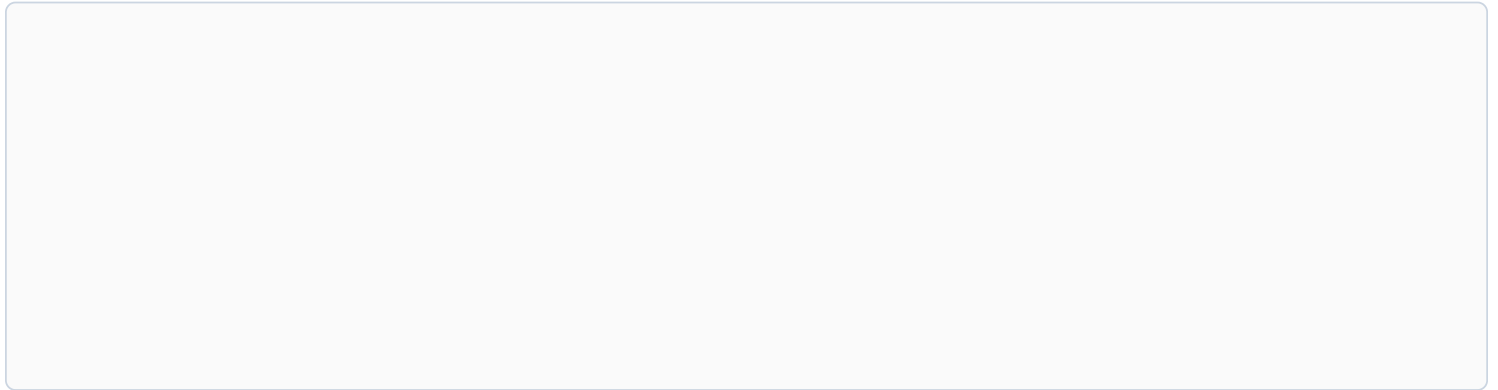
3.3 Planning and Scheduling

3.3.1 Gantt Chart (Planned vs Actual)

Phase & Description	Planned Timeline	Actual Timeline	Status
Phase 1: Requirements Gathering	Week 1 - 2	Week 1 - 2	Complete
Phase 2: UI/UX & Database Design	Week 2 - 3	Week 2 - 3	Complete
Phase 3: Setup Next.js & Clerk	Week 4	Week 4	Complete
Phase 4: Database Setup (MongoDB)	Week 5	Week 5 - 6	Complete
Phase 5: Develop Interactive Tools	Week 6 - 8	Week 6 - 8	Complete
Phase 6: Admin Dashboard Core	Week 8 - 9	Week 8 - 9	Complete
Phase 7: Twilio SMS Integration	Week 9 - 10	Week 9 - 10	Complete
Phase 8: Testing & Bug Fixes	Week 11	Week 11 - 12	Complete
Phase 9: Report & Documentation	Week 12	Week 12	Complete

3.4 Conceptual Models

3.4.1 ER (Entity-Relationship) Diagram



3.4.2 Data Dictionary

Collection: appointments

Field Name	Data Type	Key Type	Nullable	Description
_id	ObjectId	PK	No	Unique identifier generated by MongoDB.
userId	String	FK	Yes	Associates appointment with Clerk user profile (null for guests).
name	String	-	No	Patient's first and last name.
email	String	-	No	Contact email address for records.
phone	String	-	No	Recipient phone number for SMS triggers.
service	String	-	No	Type of service requested (e.g., Cleanings, Fillings).
status	String	-	No	Status tag: pending, confirmed, cancelled, etc.
scheduledDate	String/Date	-	Yes	Confirmed date assigned by the dentist/admin.
scheduledTime	String	-	Yes	Confirmed time slot assigned by the dentist/admin.
doctorNotes	String	-	Yes	Custom comments/instructions left by the dentist.
message	String	-	Yes	Additional notes typed by patient at checkout.
createdAt	Date	-	No	System timestamp when booking record was created.

3.4.3 Data Flow Diagram (DFD)

Level 0: Context Diagram

graph TD

```

Patient[Patient / Web User] →|1. Booking Request / Quiz Inputs| System[Sheetal Dental Clinic Platform]
System →|2. Form Validations & Estimated Pricing| Patient
Admin[Clinic Administrator] →|3. Reschedule & Status Updates| System
System →|4. Appointments List & Schedule Logs| Admin
System →|5. Forward Text Payload| Twilio[Twilio SMS Service]
Twilio →|6. Dispatch SMS Alerts| Patient
  
```

Level 1: Functional DFD

graph TD

```

Patient[Patient] →|Submit Form| P1[1.0 Online Intake & Booking]
P1 →|Insert Document| DB[(MongoDB - Sheetal Dental Clinic Database)]
P1 →|Register Identity| Clerk[Clerk Auth Store]

DB →|Fetch Log Entries| P2[2.0 Admin Appointment Scheduler]
Admin[Admin Dentist] →|Confirm / Reschedule / Cancel / Notes| P2
P2 →|Update Document| DB

DB →|Trigger Alerts| P3[3.0 Notification Engine]
P3 →|Trigger Endpoint Request| Twilio[Twilio SMS Gateway]
Twilio →|Transmit Text SMS| Patient

Patient →|Use Tools| P4[4.0 Assessment Suite]
P4 →|Process Inputs| P4
P4 →|Show Pricing / Treatments| Patient
  
```

3.4.4 Use Case Diagram

```
graph TB
    subgraph Sheetal_Dental_Clinic_Platform [Sheetal Dental Clinic Platform]
        UC1[UC1[Browse Services & Pricing]]
        UC2[UC2[Complete Smile Quiz & Slider]]
        UC3[UC3[Sign In via Clerk]]
        UC4[UC4[Submit Appointment Booking]]
        UC5[UC5[Manage Personal Dashboard]]
        UC6[UC6[Request Cancellation / Rescheduling]]

        UC7[UC7[View Clinic Bookings list]]
        UC8[UC8[Approve / Cancel Appointments]]
        UC9[UC9[Assign Schedule Date, Time & Doctor Notes]]
        UC10[UC10[Switch Access Roles]]
    end

    Patient([Patient / User]) --> UC1
    Patient --> UC2
    Patient --> UC3
    Patient --> UC4
    Patient --> UC5
    Patient --> UC6

    Admin([Clinic Administrator]) --> UC3
    Admin --> UC7
    Admin --> UC8
    Admin --> UC9
    Admin --> UC10
```

3.4.5 Activity Diagram

graph TD

Start([Start]) → Browse[Browse Site & Tools]

Browse → Auth{Authenticated?}

Auth →|No| Login[Log in / Register via Clerk]

Login → Auth

Auth →|Yes| Booking[Fill Appointment Form]

Booking → Submit[Submit to Server]

Submit → DBInsert[Insert Record into MongoDB]

DBInsert → SMSRequest[Send Twilio Alert: Request Received]

SMSRequest → AdminDashboard[Admin Reviews in Control Panel]

AdminDashboard → Decision{Action Decision}

Decision →|Approve Booking| Confirmed[Set status: confirmed & enter schedule metadata]

Decision →|Reject Booking| Cancelled[Set status: cancelled]

Confirmed → DBUpdate[Update MongoDB Record]

Cancelled → DBUpdate

DBUpdate → SMSTrigger[Dispatch Twilio Update Notification to Patient]

SMSTrigger → End([End])